

Funzioni

Siano X, Y insiemi non vuoti.

Una funzione f da X a Y ($f: X \rightarrow Y$) è una legge che associa ad ogni elemento $x \in X$ uno e un solo elemento $y \in Y$.

↓
argomento
della funzione

↓
valore
della funzione
in x , $f(x) = y$

$$f: X \rightarrow Y \longrightarrow \text{codominio}$$

$Y \subseteq \mathbb{R}$

↪ dominio $X \subseteq \mathbb{R}$

$$\text{graf } f = \{ (x, y) \in \vec{X \times Y} : y = f(x) \}$$

$$\text{im } f = \{ y \in Y : \exists x \in X : f(x) = y \}$$

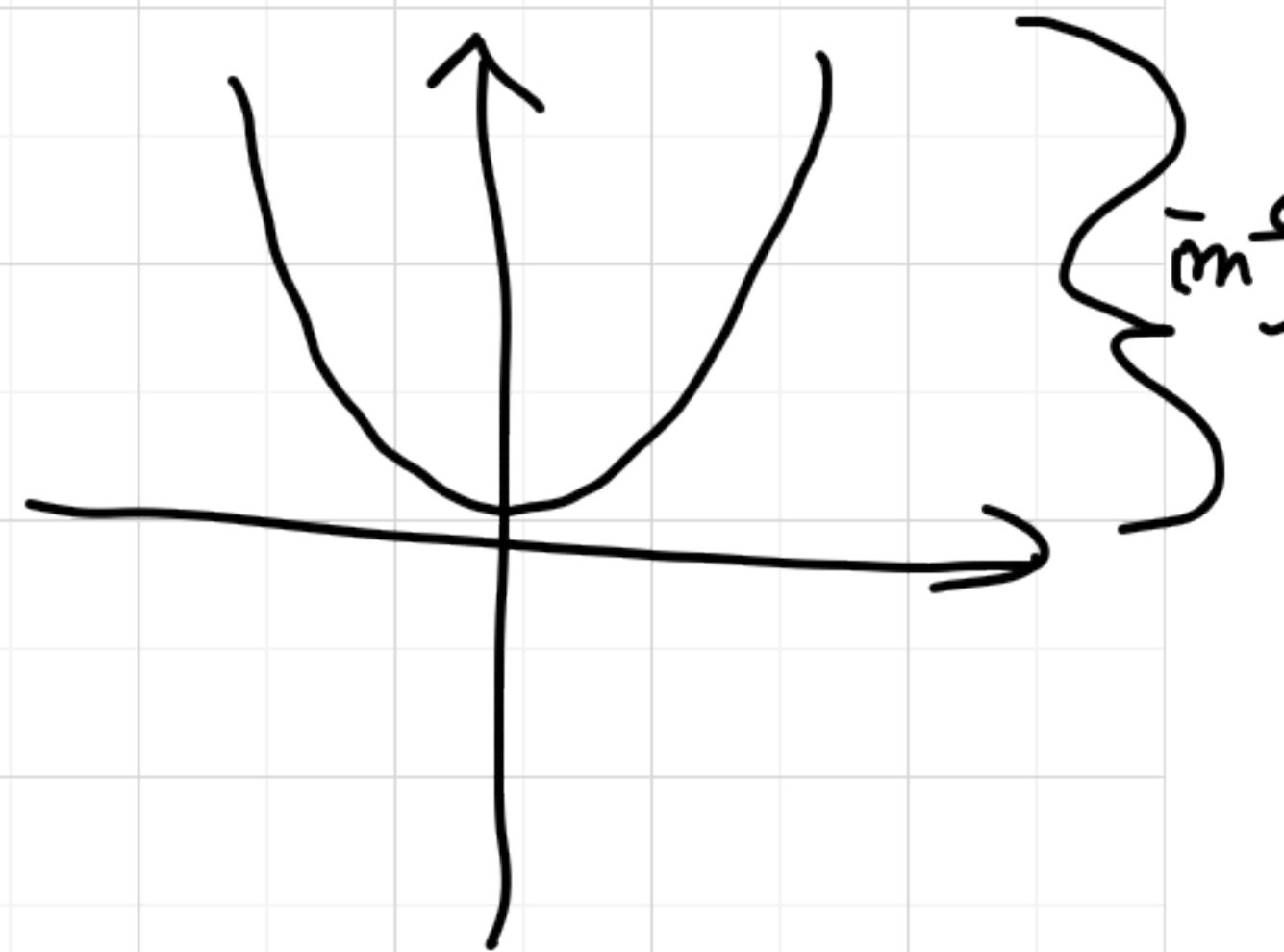
$$= \{ f(x) : x \in X \}$$

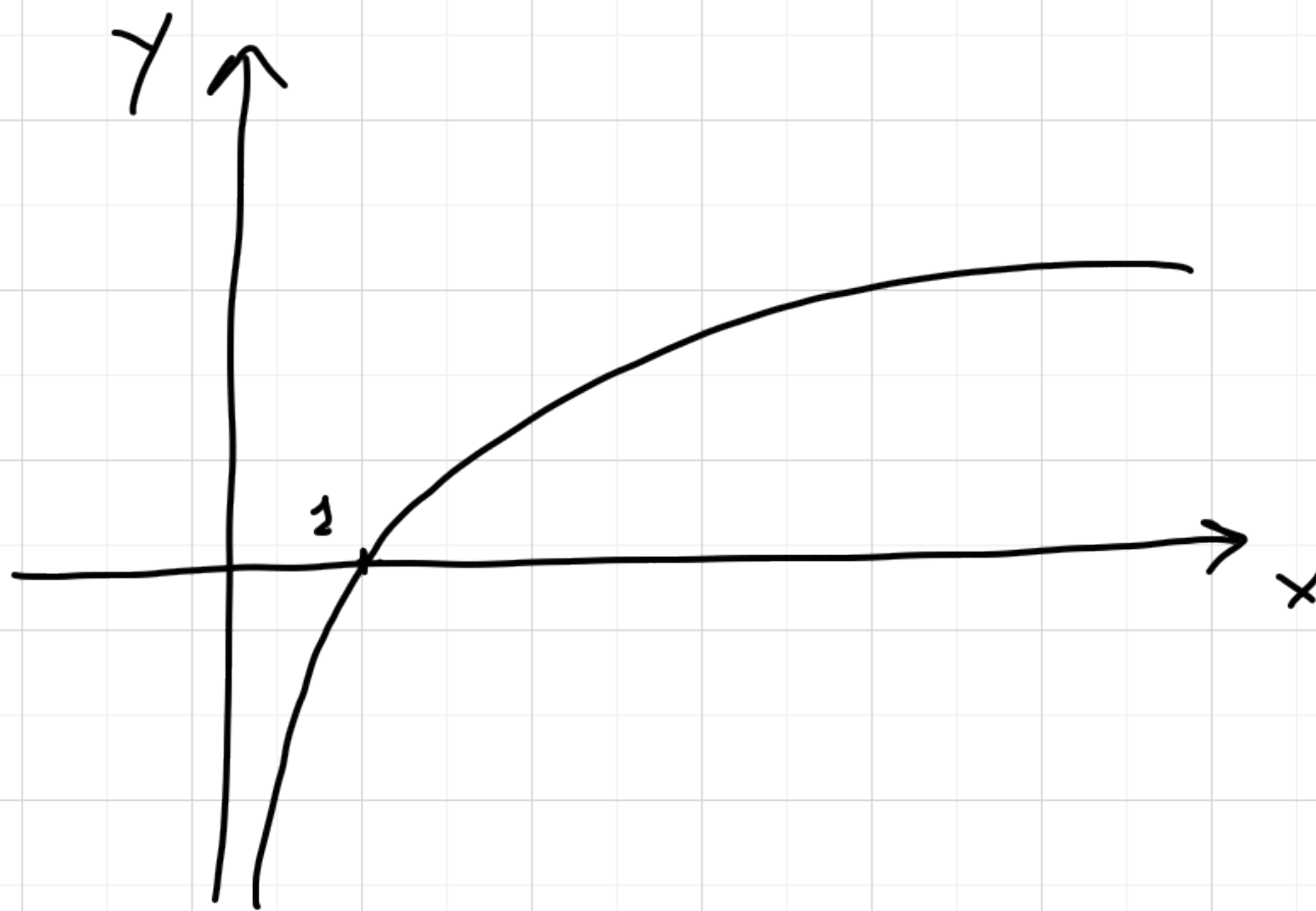
$$Y = \mathbb{R}$$

↓
codominio

$$\text{im } f \subseteq \text{codom } f$$

$$f: \mathbb{R} \rightarrow \mathbb{R} \quad y = x^2$$





$$y = \log_a(x)$$

$$X = (0, +\infty)$$

$$\text{im } f = \text{codom} = \mathbb{R}$$

$$f = (0, +\infty) \rightarrow \mathbb{R}$$

$$X = [0, +\infty)$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$y = f(x) = \frac{1}{1+x^2}$$

$$\hookrightarrow 1+x^2 \geq 1$$

$$x < 0$$

$$f(x) = \frac{1}{1+(-\infty)^2} = \frac{1}{\infty}$$

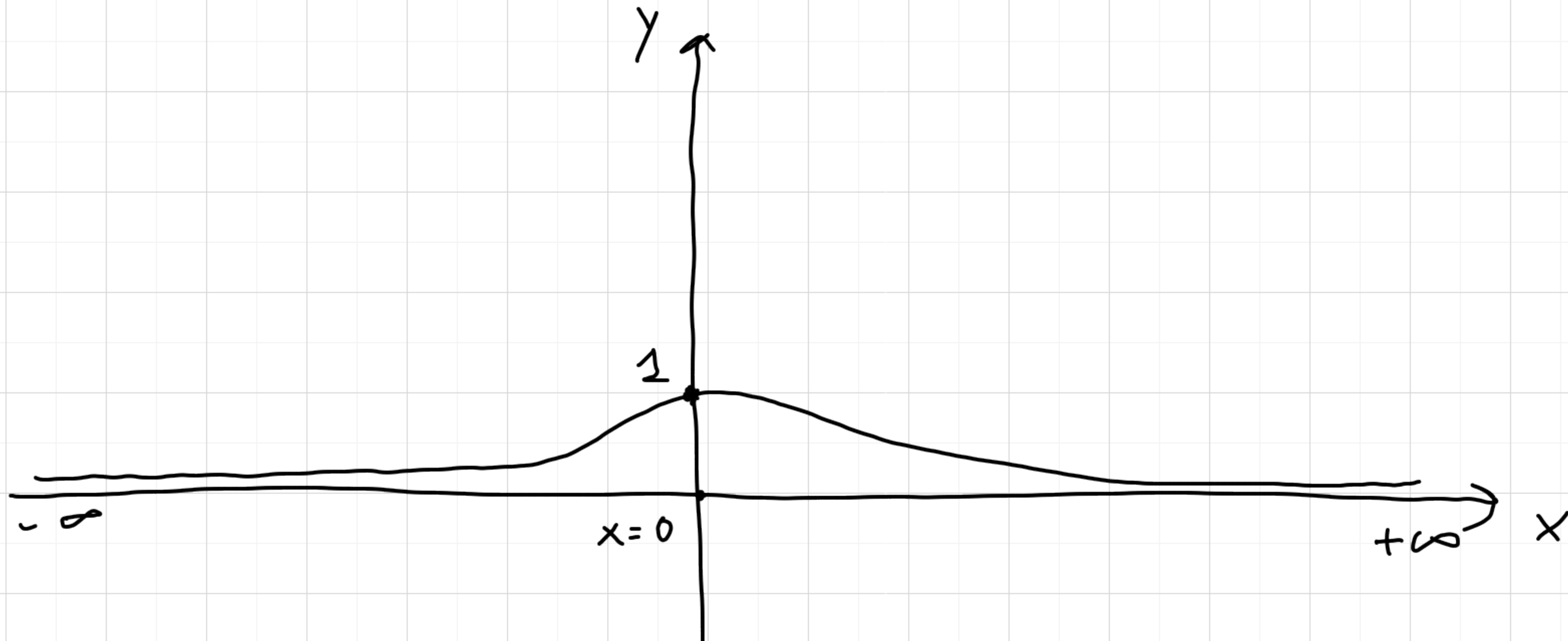
$$x > 0$$

$$f(x) = \frac{1}{1+(+\infty)^2} = \frac{1}{\infty}$$

$$x = 0$$

$$f(0) = \frac{1}{1+0} = 1$$

$$\text{Im } f = (0, 1]$$



Uguaglianza

Una funzione $f = X \rightarrow Y$, una funzione $g = X' \rightarrow Y'$.

f e g sono uguali se:

i) $X = X'$ (hanno lo stesso dominio)

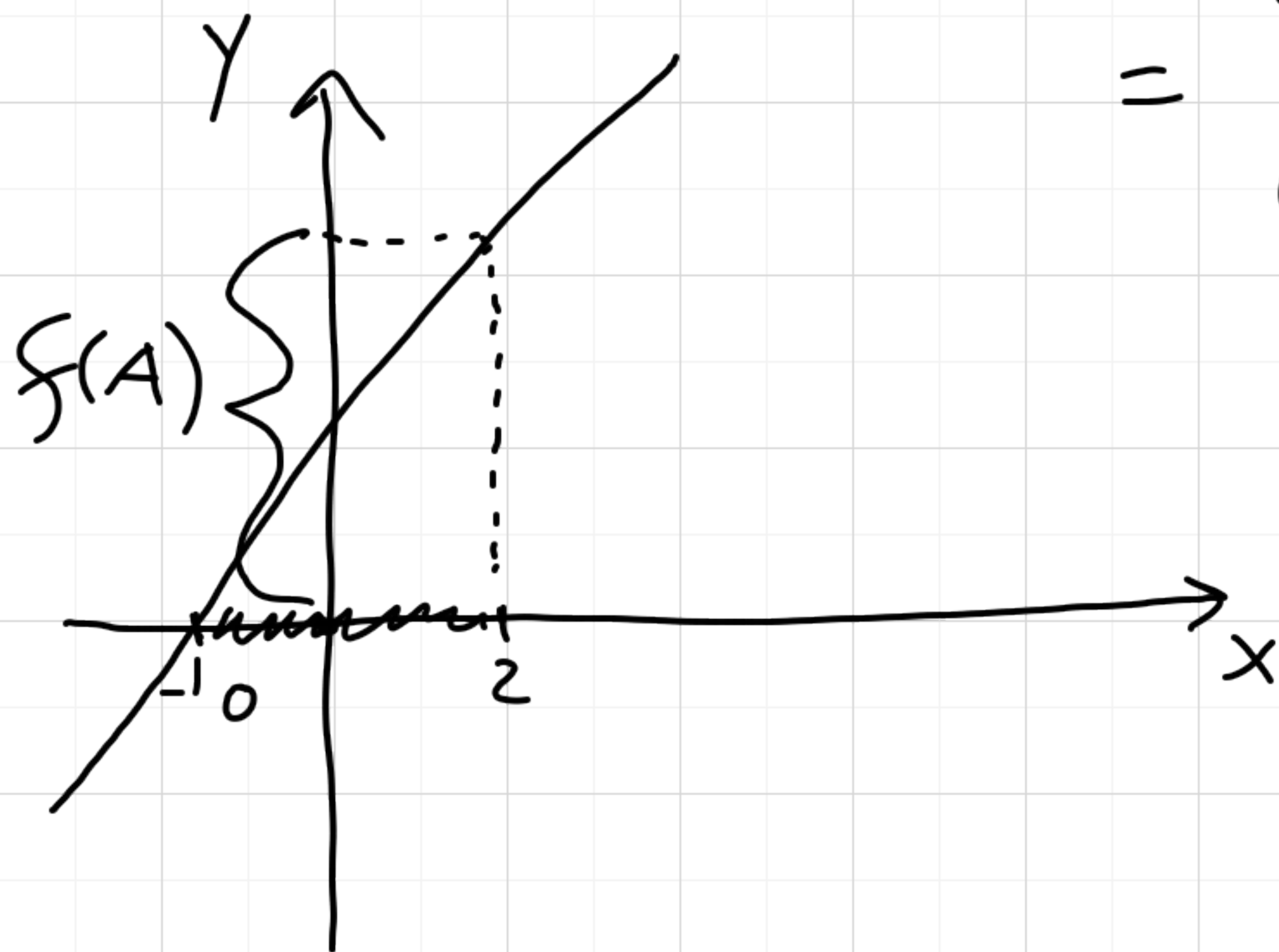
ii) $f(x) = g(x) \quad \forall x \in X$ (stesso valore funzione)

Immagine e controimmagine

Sia $f: X \rightarrow Y$ e $A \subseteq X$ - l'immagine di A

è l'insieme $f(A) = \{f(x) : x \in A\} =$

$$= \{y \in Y : \exists x \in A : y = f(x)\}$$

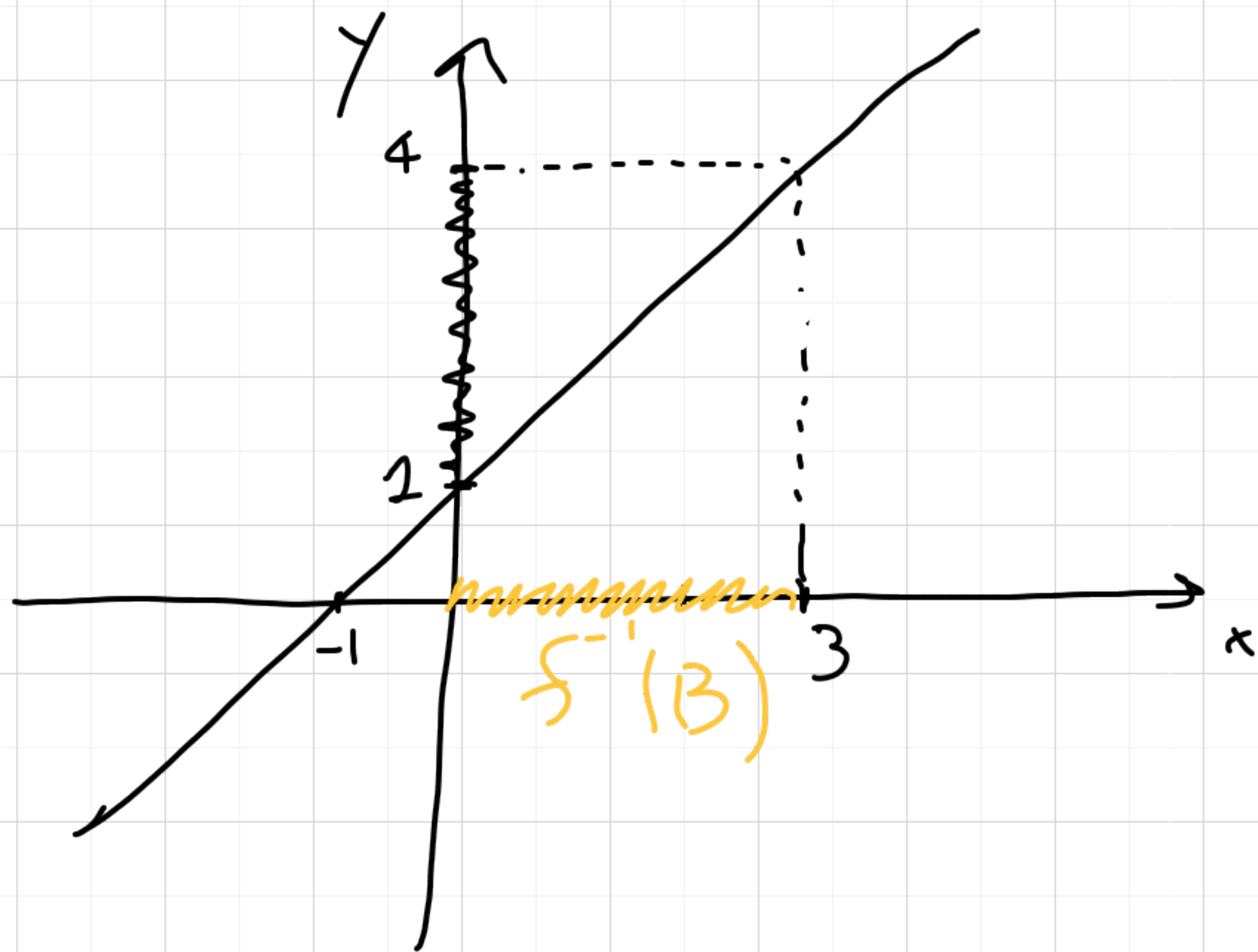


$$A = [-1, 2]$$

$$\begin{aligned} A &\subseteq X \\ f(A) &\subseteq Y \end{aligned}$$

l'insieme $B \subseteq Y$ si dice contrainmagine di B e si indica

$$f^{-1}(B) = \{x \in X : f(x) \in B\}$$



$$B = [1, 4]$$

$$\begin{array}{l} B \subseteq Y \\ f^{-1}(B) \subseteq X \end{array}$$

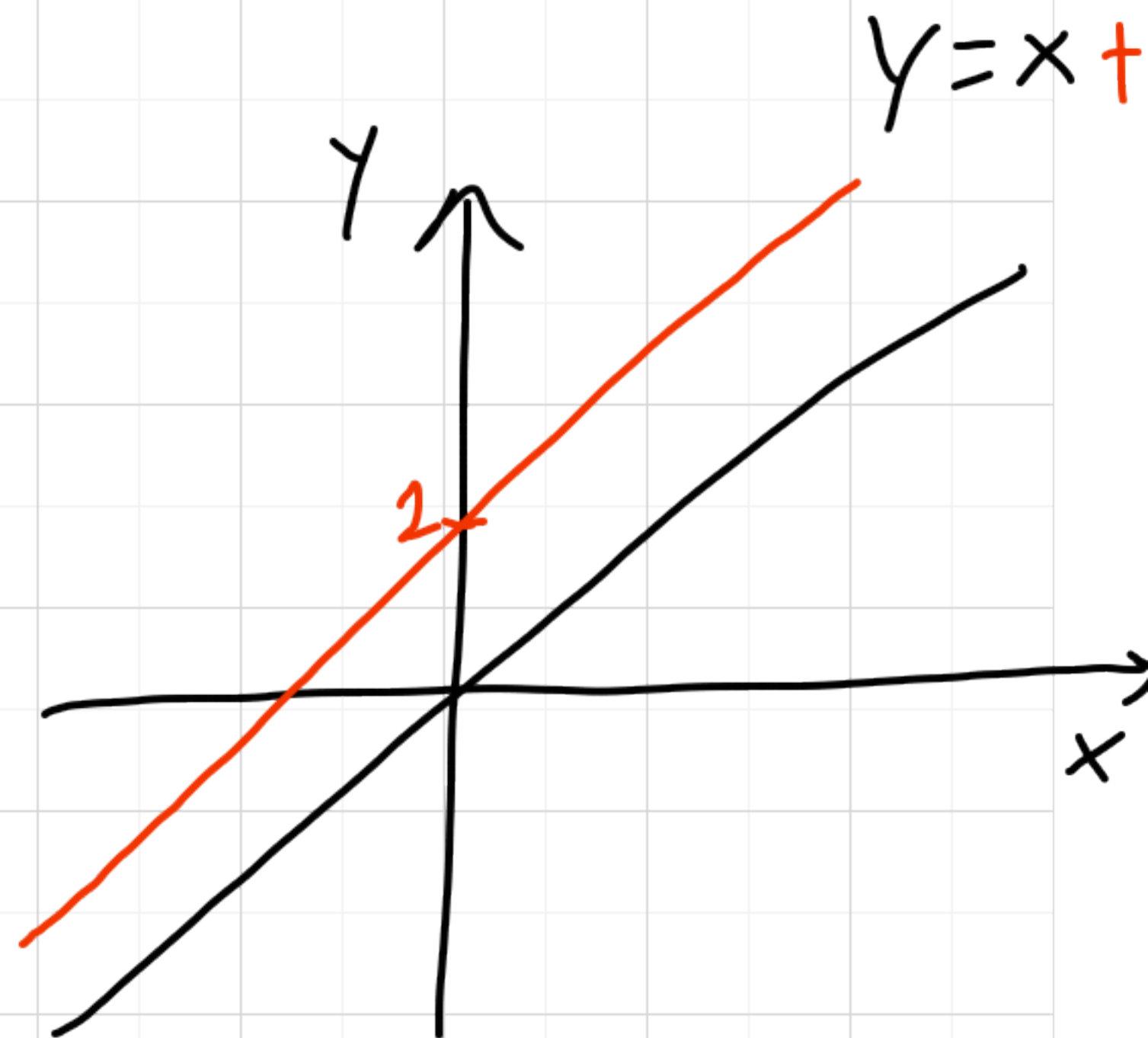
Operazioni di Funzioni

Sia $X \subseteq \mathbb{R}$, $f: X \rightarrow \mathbb{R}$

① $k \in \mathbb{R} \setminus \{0\}$ TRASLAZIONE RISPETTO A y

$$g(x) = f(x) + k$$

- se $k > 0 \Rightarrow$ TRASLO $\uparrow$
- se $k < 0 \Rightarrow$ TRASLO $\downarrow$



Dominio =
Immagine $\neq$

② $x_0 \in \mathbb{R} \setminus \{0\}$: TRASLAZIONE RISPETTO A X

$$g(x) = f(x + x_0)$$

- Se $x_0 > 0 \Rightarrow$ TRASLO $\leftarrow$

- Se $x_0 < 0 \Rightarrow$ TRASLO $\rightarrow$

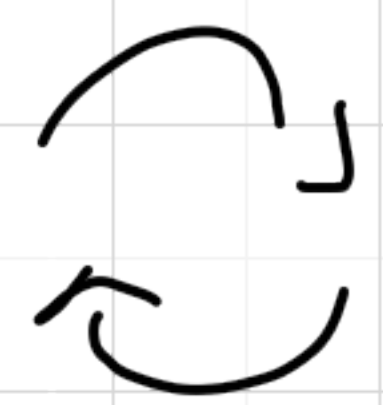
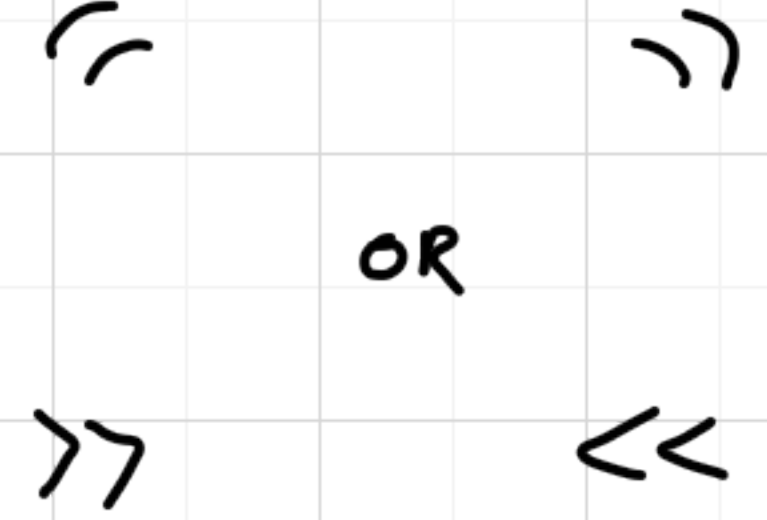
DOMINIO $\neq$

IMMAGINE $=$

③ $A \in \mathbb{R} \setminus \{0\}$ = RISCALAMENTO RISPETTO A Y

$$g(x) = A f(x)$$

verticalmente

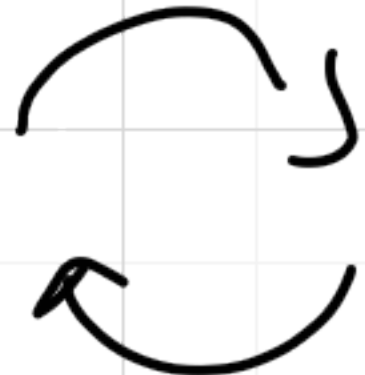
- se $A > 1 \Rightarrow$ "ESPANSIONE"
- se $0 < A < 1 \Rightarrow$ "COMPRESSIONE"
- se $A < 0 \Rightarrow$  + "OR" 

④ $B \in \mathbb{R} \setminus \{0\}$: RISCALAMENTO RISPETTO A X

$$g(x) = f(Bx)$$

orizzontale

$\left\{ \begin{array}{l} - \text{ se } B > 1 \Rightarrow \gg \text{ COMPRESSIONE } \ll \\ - \text{ se } 0 < B < 1 \Rightarrow \hat{=} \text{ ESPANSIONE } \hat{=} \end{array} \right.$

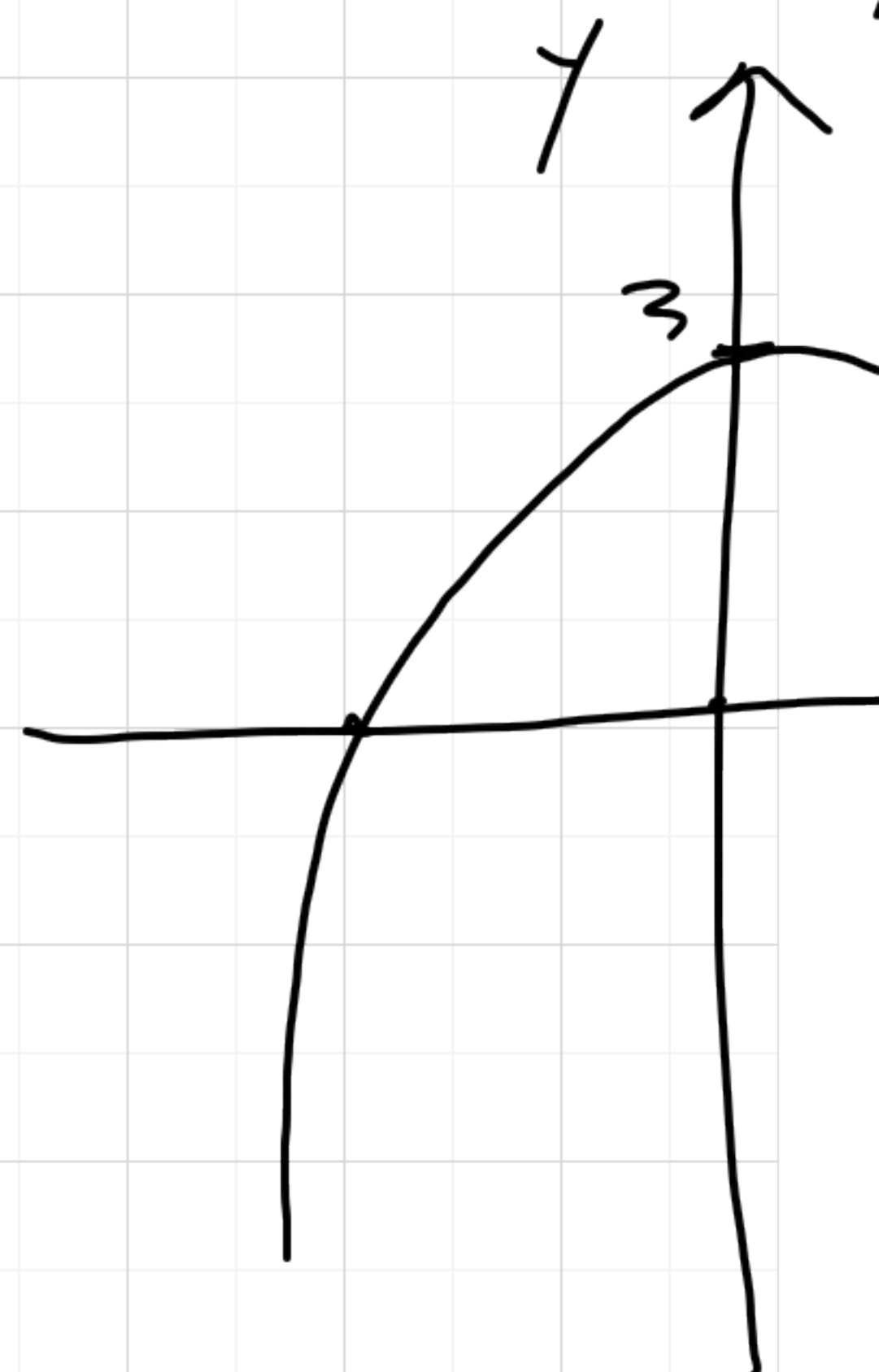
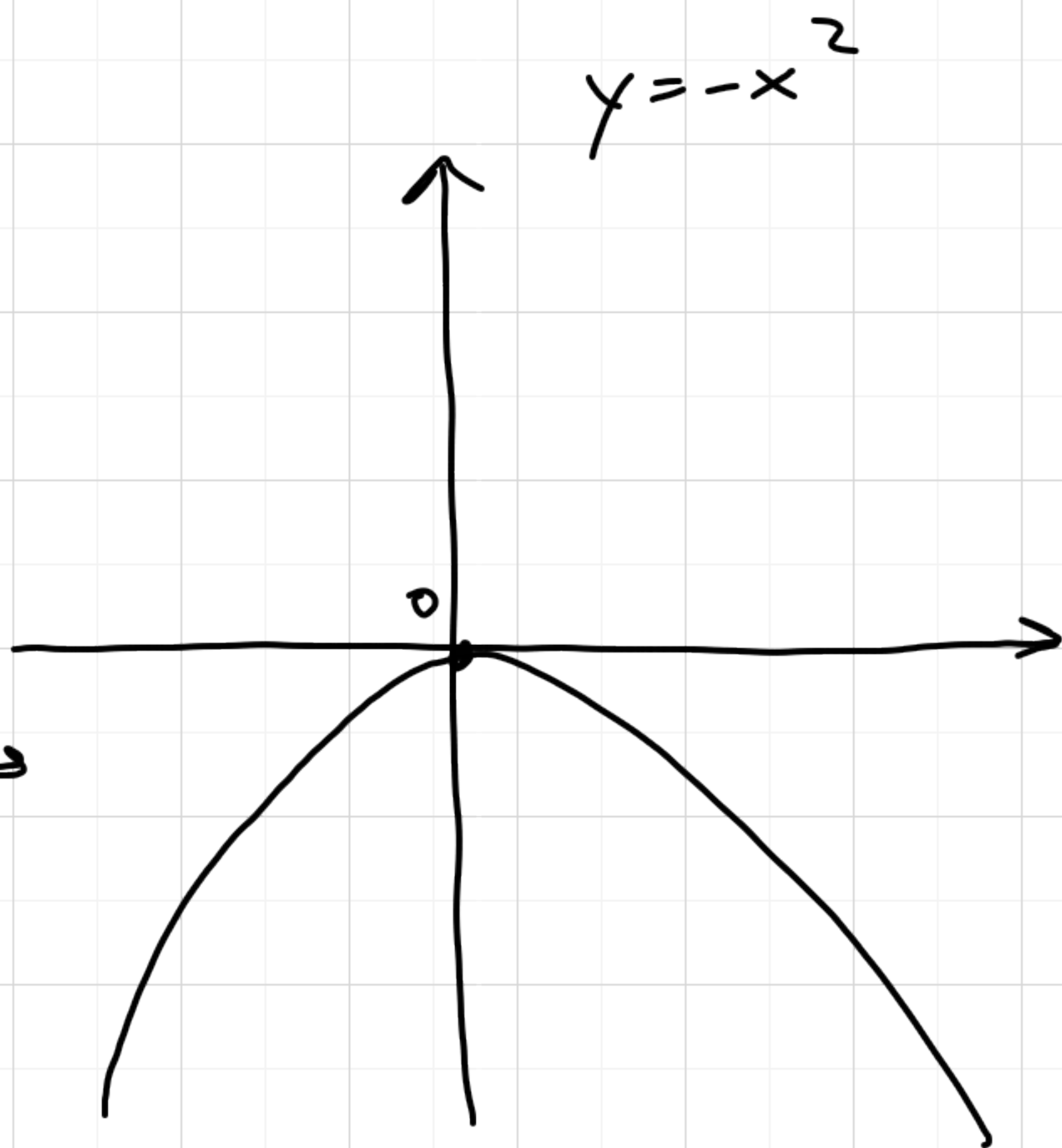
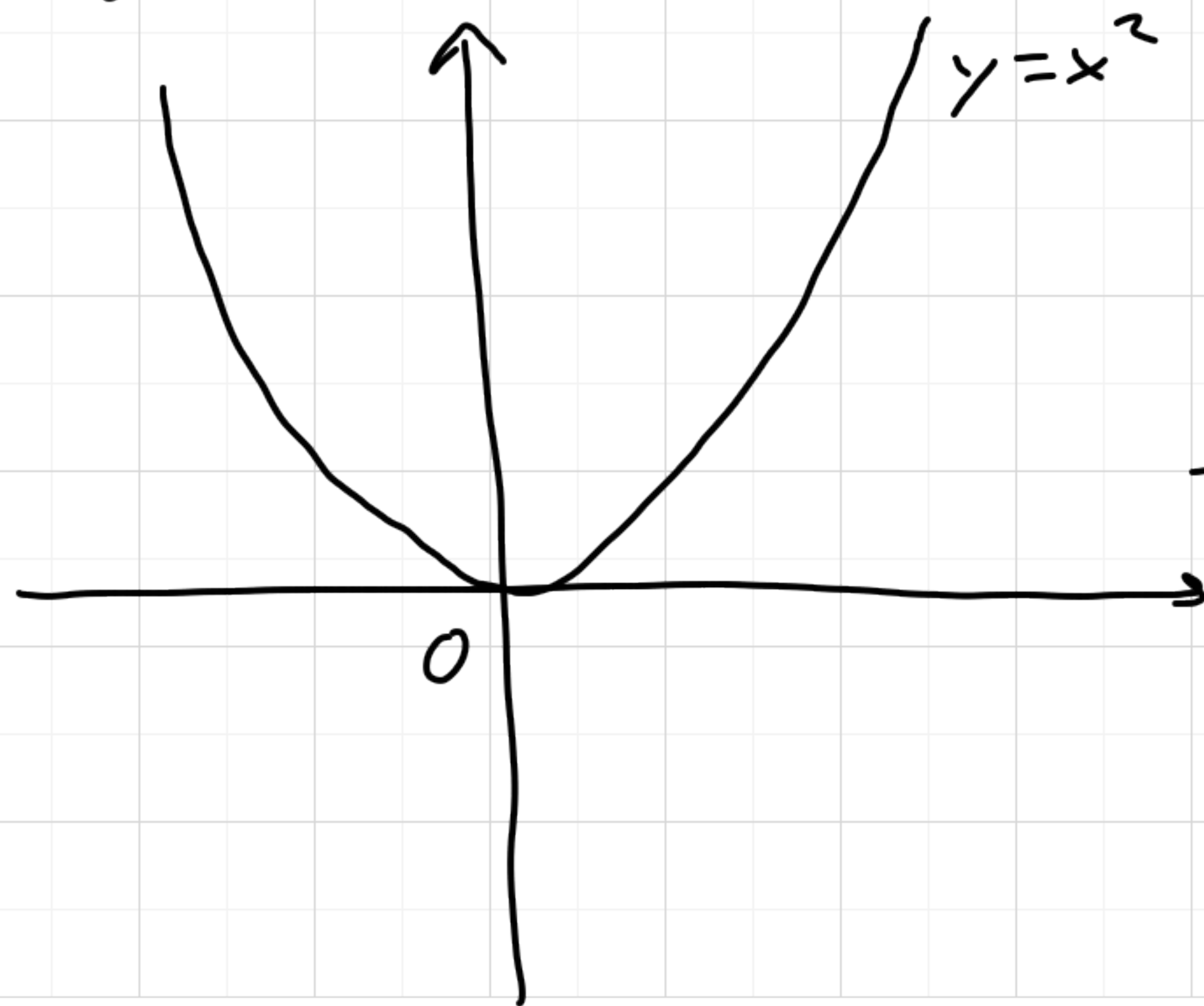
$- \text{ se } B < 0 \Rightarrow$  $+ \gg \text{ or } \ll$

Esempio

$$g(x) = 3 - x^2 = -x^2 + 3$$

$$g(x) = (-1)x^2 + 3$$

$$f(x) = x^2$$



Parte positiva e negativa

$$\text{Parte } + = f_+ = \begin{cases} f(x) & \text{se } f(x) > 0 \\ 0 & \text{se } f(x) \leq 0 \end{cases}$$

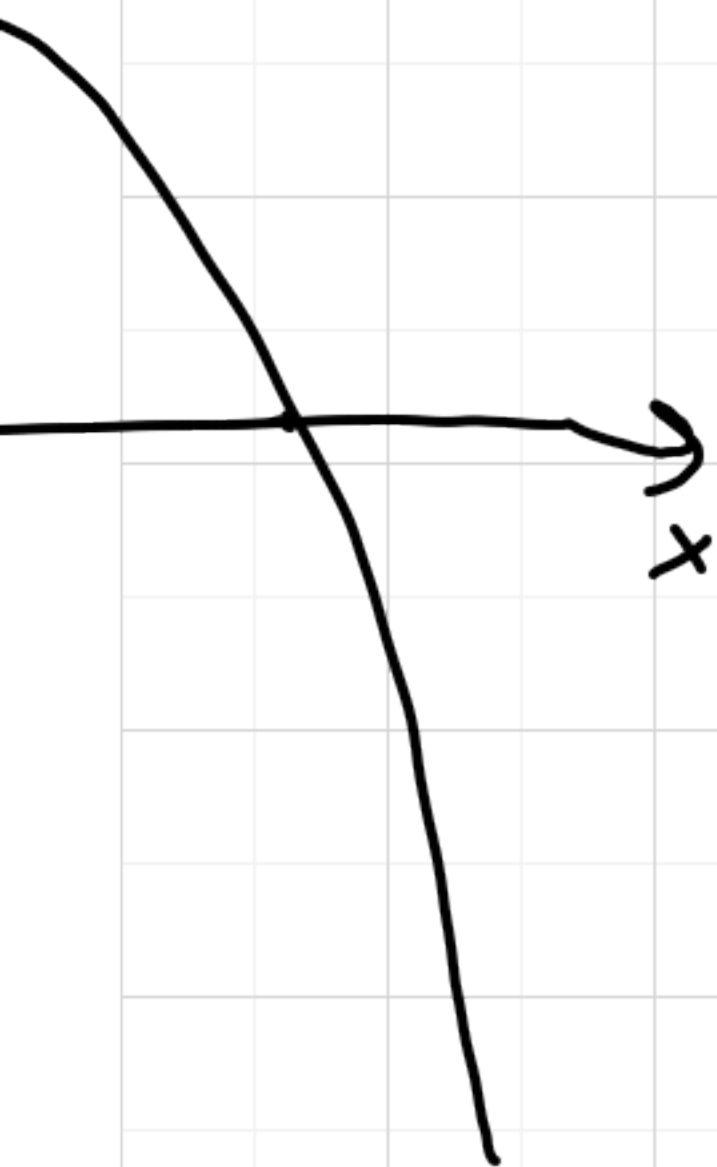
$$\text{Parte } - = f_- = \begin{cases} 0 & \text{se } f(x) \geq 0 \\ -f(x) & \text{se } f(x) < 0 \end{cases}$$

NOTA: entrambe le funzioni sono sempre non negative

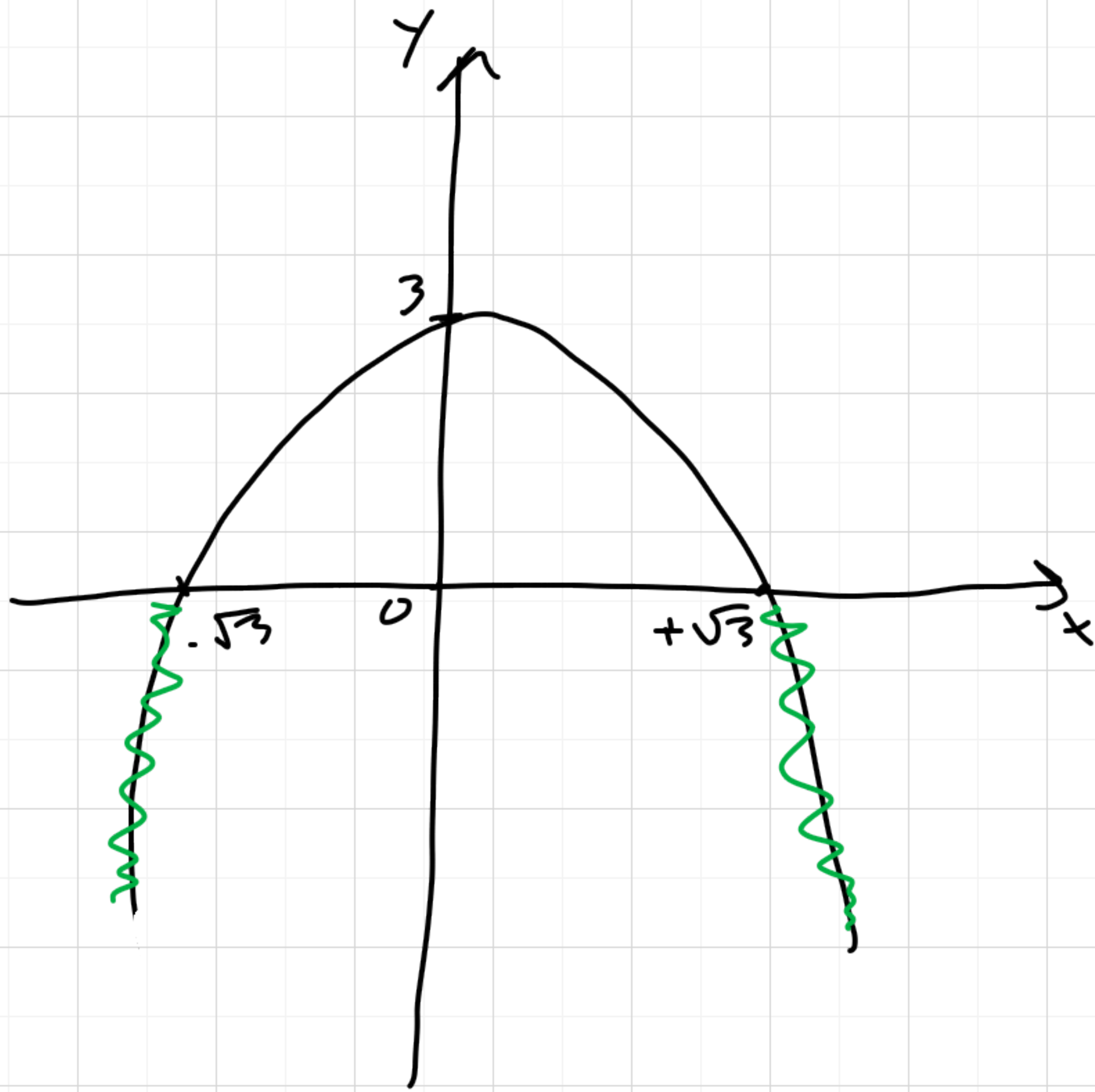
$$f_+ = \max \{0, f(x)\}$$

$$f_- = \max \{0, -f(x)\}$$

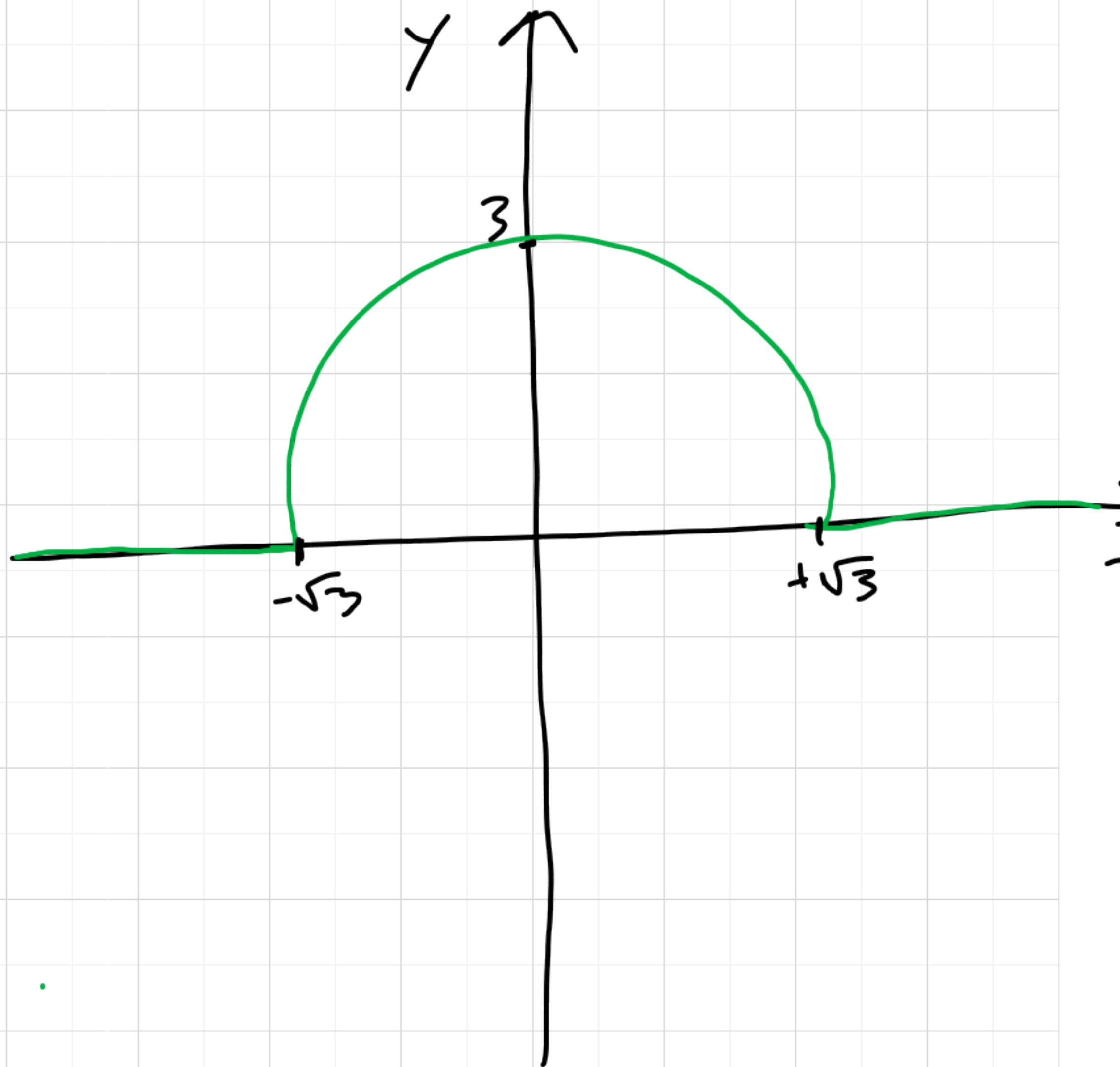
$$y = -x^2 + 3$$



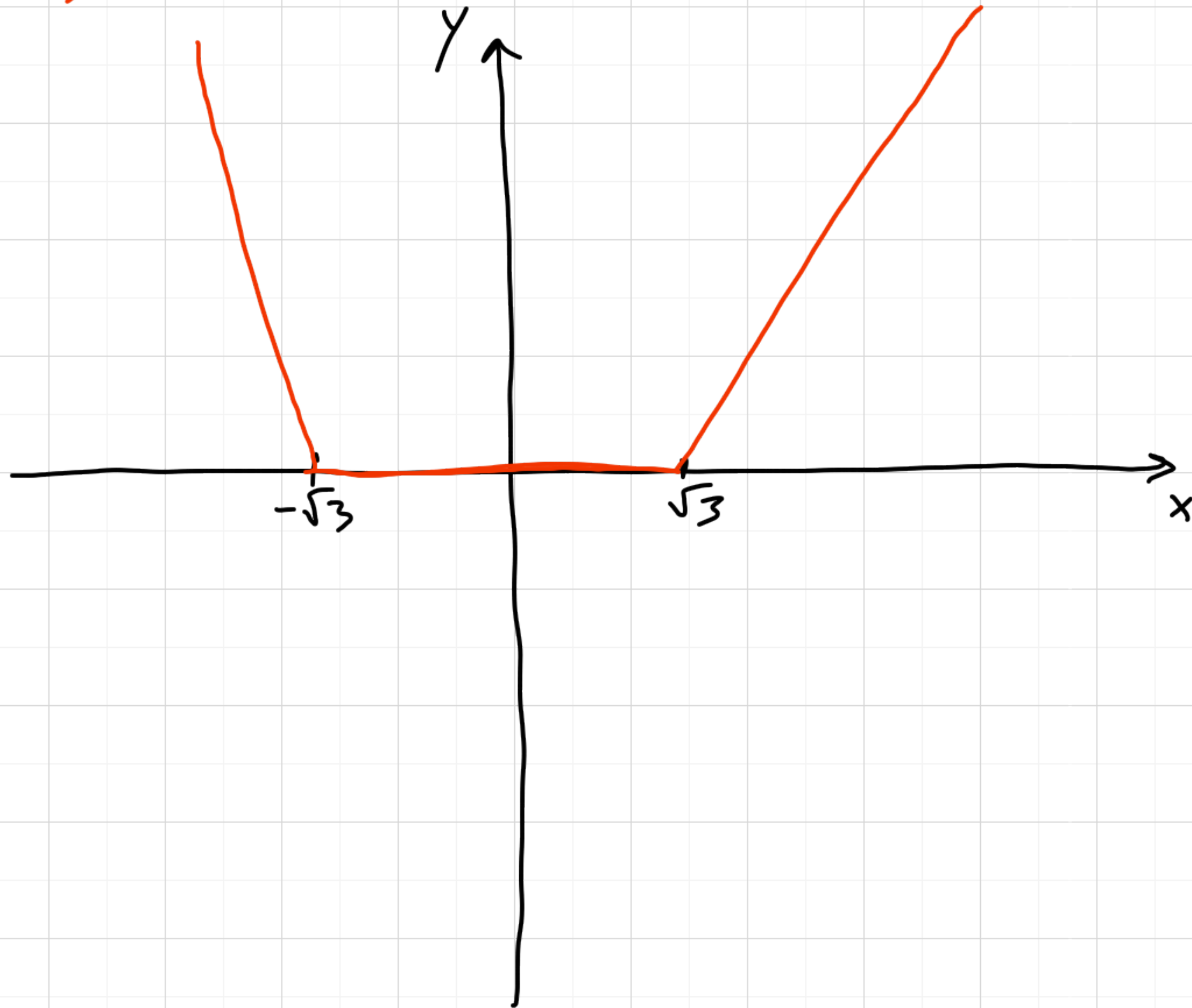
$$f(x) = 3 - x^2$$



$$f_+(x) = \max\{0, f(x)\}$$

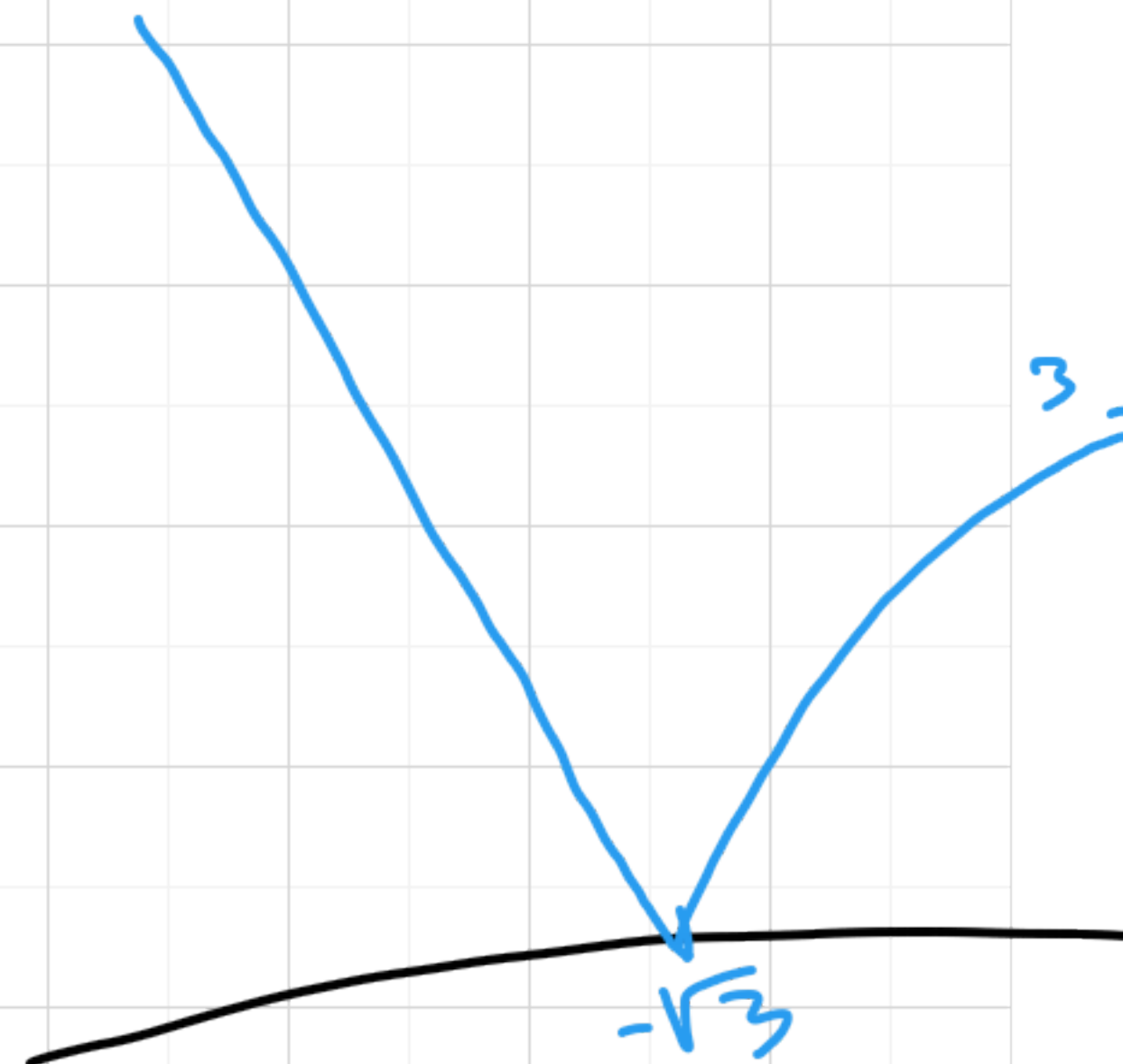


$$f_-(x) = \max \{ 0, -f(x) \}$$

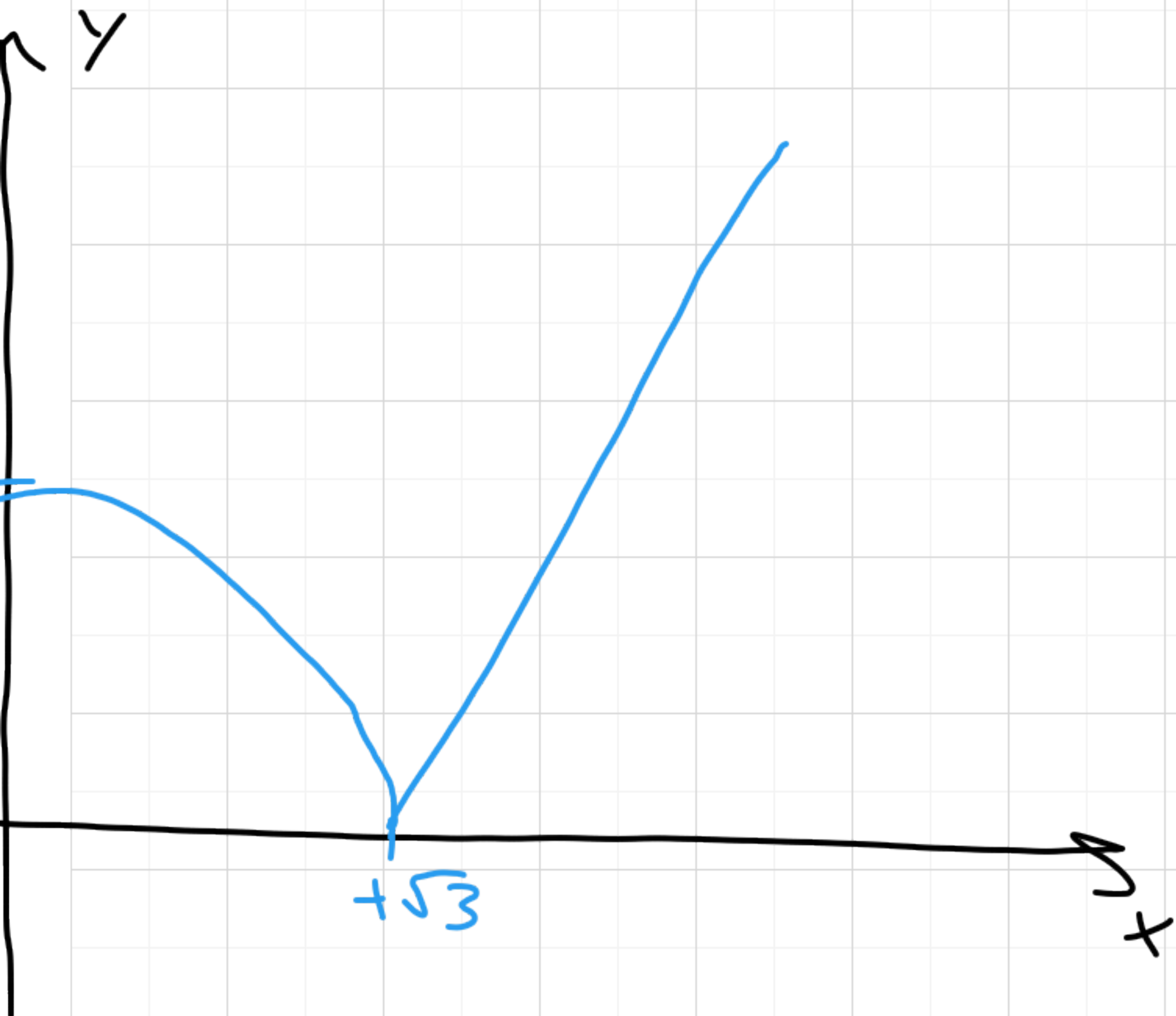


$$|f(x)| = f_+(x) - f_-(x)$$

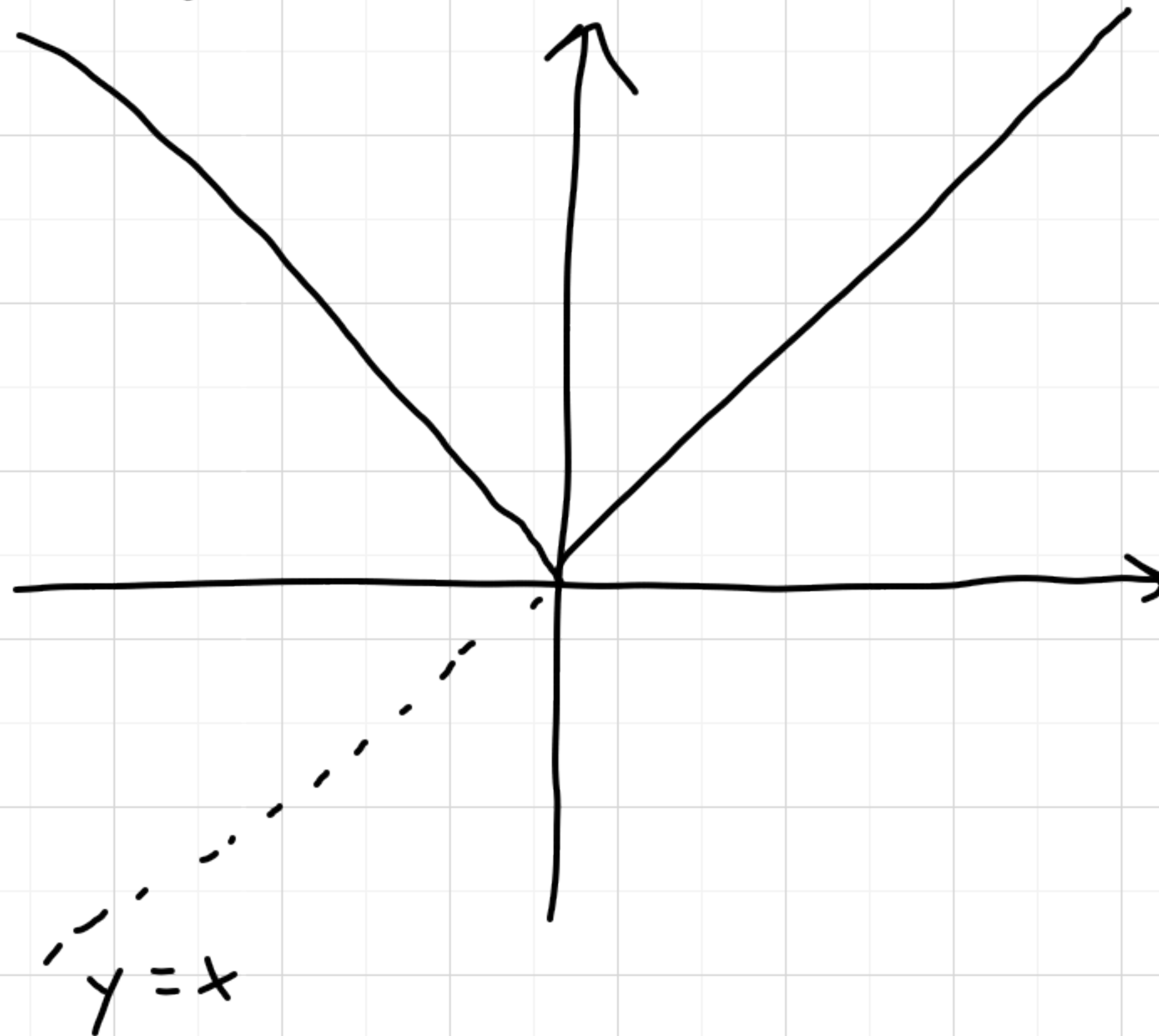
$$f(x) = |3 - x^2|$$



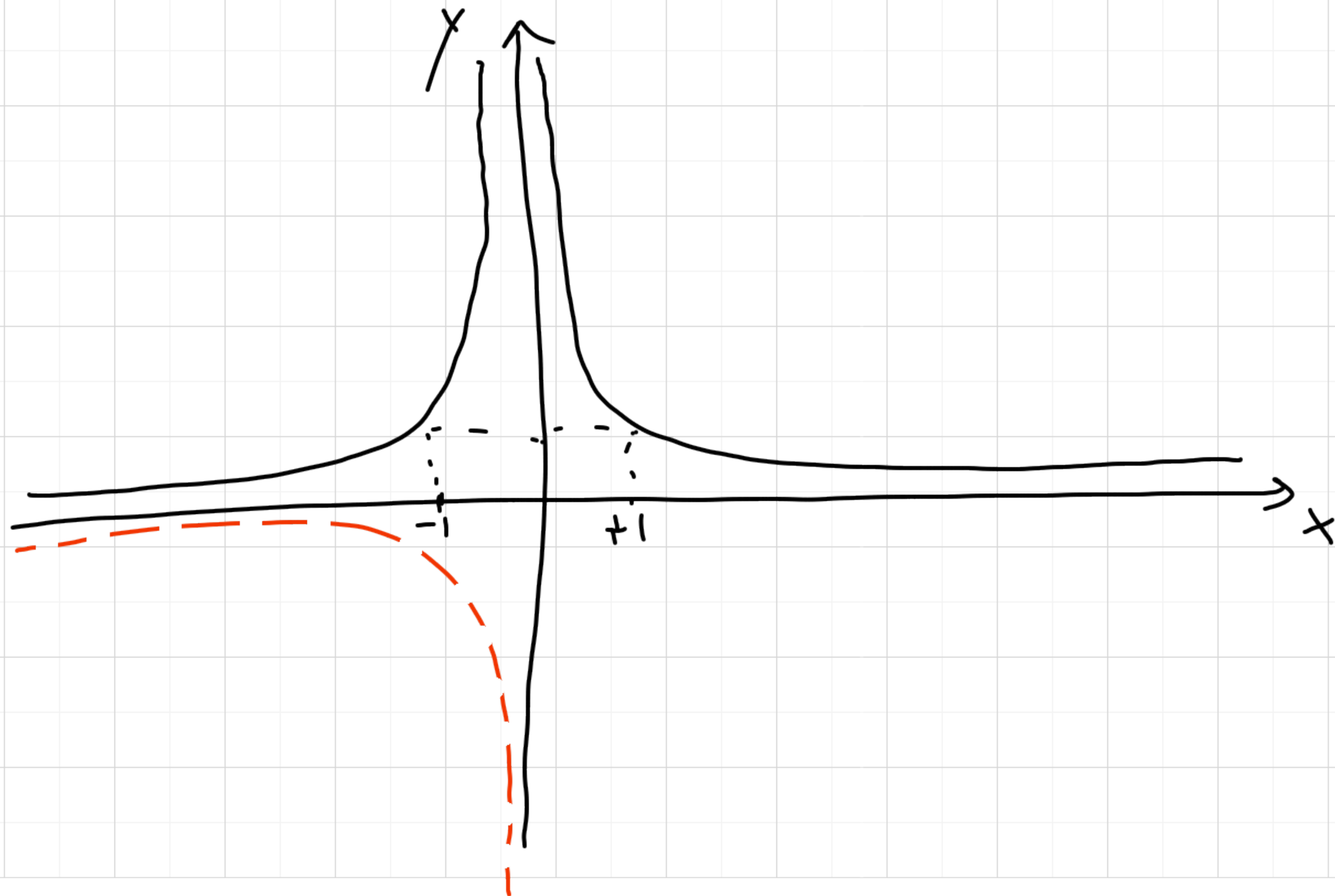
$f_-(x)$



$|x|$



$$g(x) = \frac{1}{|x|} = \frac{|1|}{|x|} = \left| \frac{1}{x} \right|$$



$$y = \frac{1}{x}$$

Funzione periodiche

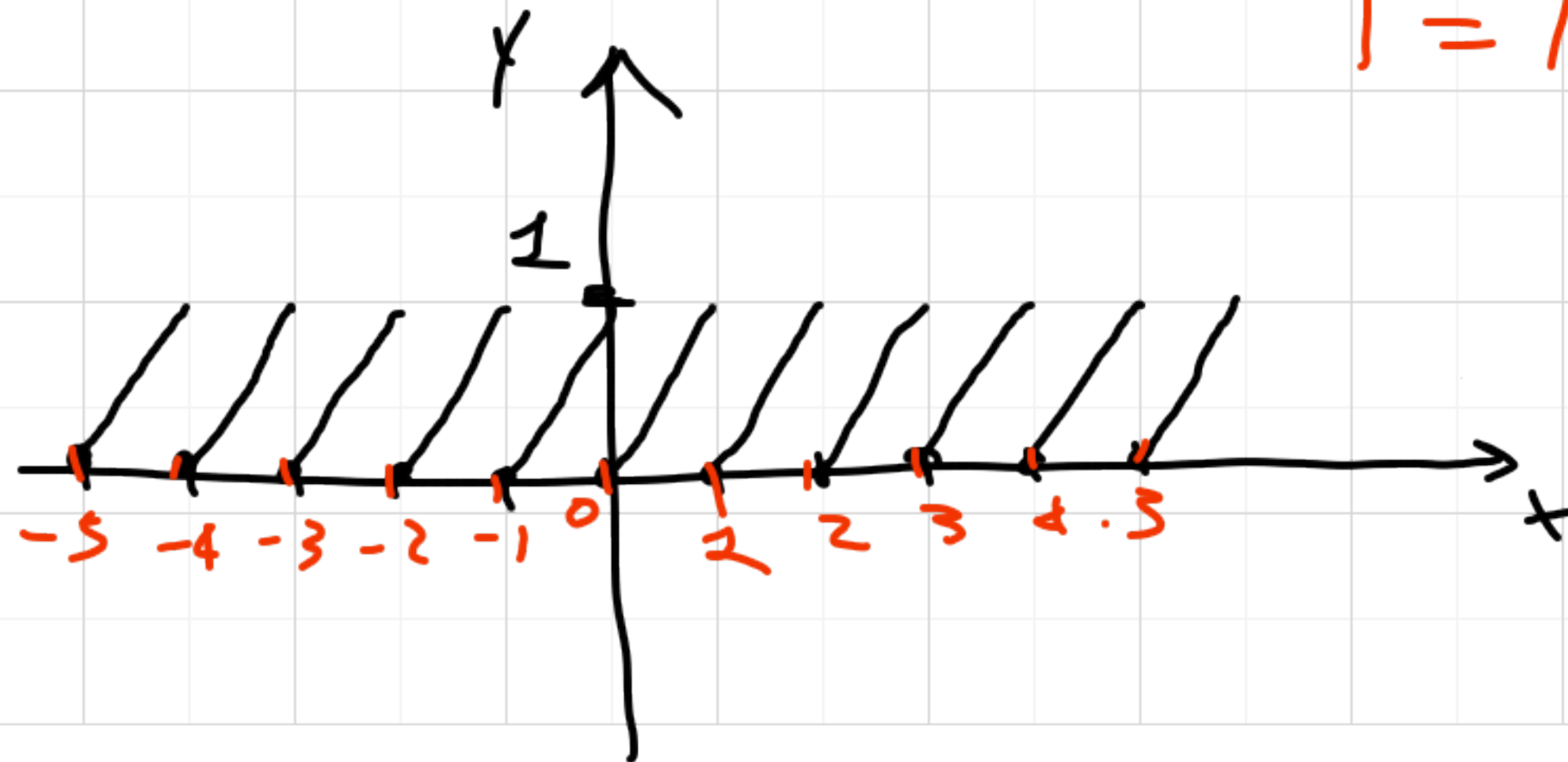
Sia $X \subseteq \mathbb{R}$, $f: X \rightarrow \mathbb{R}$ si dice periodica di periodo $T > 0$

se

$$\begin{cases} x+T \in X & \forall x \in X \\ f(x+T) = f(x) & \forall x \in X \end{cases}$$

periodo
fondamentale

$$f(x) = x - \lfloor x \rfloor$$



$$x = 1,15$$

$$f(1,15) = 1,15 - 1 = 0,15$$

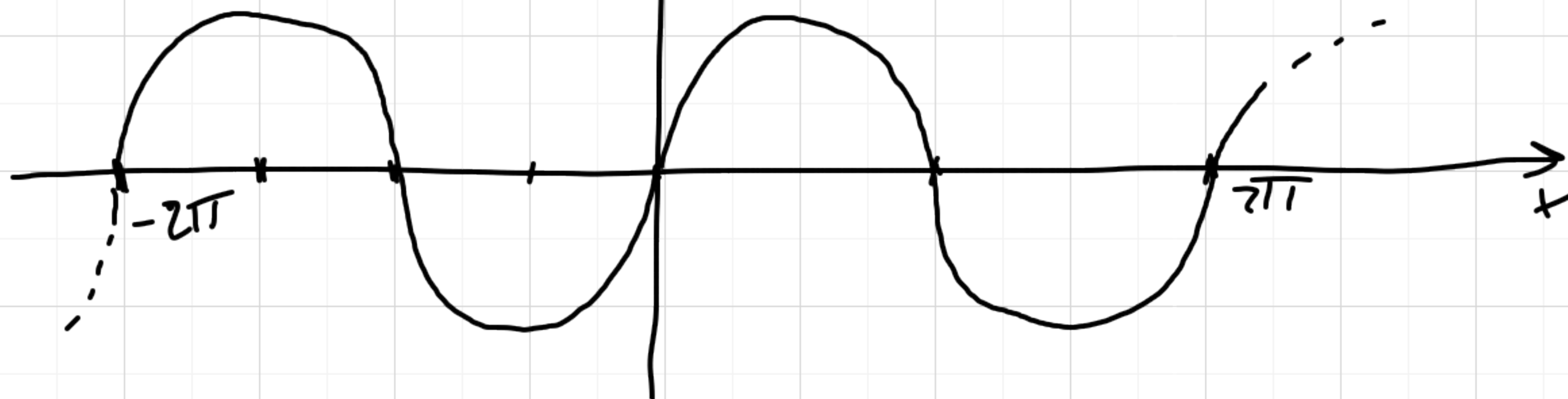
$$f(2) = 2 - 2 = 0$$

$$f(1) = 1 - 1 = 0$$

$$f(1,0001) = 1,0001 - 1 = 0,0$$

$$y = \sin x$$

$$T = 2\pi$$



$$y = \cos x$$

